Fiscal Year:	FY 2023	**Task Last Updated:**	FY 03/31/2024
PI Name:	Blanchard, Joel W Ph.D.		
Project Title:	Identification of Biomarkers and Pathological Mechanisms via Longitudinal Analysis of Neurological and Cerebrovascular Responses to Neurotoxic Stress Using a Multi-cellular Integrated Model of the Human Brain		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	10029	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	09/28/2022	**End Date:**	09/29/2026
No. of Post Docs:	3	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	No Changes		
COI Name (Institution):	Han, Woojin Ph.D. (ICAHN School Of Medicine at Mount Sinai)		
Grant/Contract No.:	80ARC022CA004		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As we age, our brains are naturally exposed to increasing amounts of neurotoxic insults, such as DNA damage, neuroinflammation, and amyloid-beta. In many, the buildup of these stressors diminishes cognition by impairing critical processes such as learning and memory. However, across the human population, cognition does not decline uniformly with age, or quantity of neurotoxic insults. Some individuals appear particularly vulnerable to stress induced cognitive impairments, while others are highly resilient. Genomic studies have identified genetic polymorphisms that correlate with accelerated cognitive decline and genetic variants that may even provide cognitive resilience. Other factors, such as lifestyle, diet, or certain prescribed drugs, also accelerate stress induced cognitive impairments. Unfortunately, our mechanistic understanding of these processes is rather limited. Much of what we know is from epidemiologic correlation studies. The vast genetic and lifestyle diversity among humans introduces numerous confounding variables that are problematic to control for, making it difficult to pinpoint causal factors and underlying mechanisms. Due to these limitations, it is nearly impossible to identify individuals during the earliest stages of cognitive degeneration and there are currently no therapeutic strategies to prevent or reverse its course. We develop induced pluripotent stem cell (iPSC)-based in vitro micro-physiological models of the human brain that incorporate CRISPR/CAS9 genome editing to precisely determine how genetic risk variants promote pathogenic phenotypes associated with cognitive decline, identify biomarkers, and develop therapeutic strategies around these insights. We recently applied our microphysiological systems (MPS) models to determine how the strongest genetic risk factor for Alzheimer's disease (AD), APOE4, causes cerebrovascular disease (Nature Med., 2020) and hypomyelination (under review at Nature). Over the past two years, we developed the multi-cellular integrated brain-on-a-chip (miBrain) as a project funded by the National Institutes of Health (NIH) National Institute of Neurological Disorders and Stroke (NINDS) (1UG3NS115064-01). The miBrain incorporates nine cell-types from the human brain (inhibitory and excitatory neurons, astrocytes, microglia, oligodendrocytes, oligodendrocyte precursor cells/OPCs, endothelial cells, pericytes, and vascular fibroblasts). Cells within the miBrain organize to form physiological tissues of the human brain, including the blood-brain barrier, neuronal circuits, myelinated axons, and neuroinflammation. We iteratively optimized and validated the miBrain via histological and transcriptomic comparison to the post-mortem human brain. We validated the miBrain can investigate disease states such as those induced by the SARs-COV2 virus and genetic AD risk factors. The majority of our studies have focused on acute neurological responses. Our preliminary studies have determined it can be cultured for at least 6 months. We are eager to apply and validate the miBrain platform further for long-term longitudinal studies. Here, we propose to apply the miBrain to longitudinally investigate neurological and cerebrovascular responses to common neurotoxic stresses commonly encountered on Earth and during spaceflight. Our central hypothesis is that genetics and drugs interact with neurotoxic stress to influence the physiological and pathological response of human brain tissue. In Aim 1, we will optimize the miBrain platform long-term longitudinal studies, integrating an automated feeding system and low cost mini-microscopes for continuous non-invasive monitoring. In Aim 2, we will construct and validate a panel of multiplexed assays for continuously monitoring miBrain response to acute and chronic exposure neurotoxic stress. In Aim 3, we will pinpoint how genetics interact with neurotoxic stress and contribute to the functions and dysfunctions of the human brain. We will also deploy the miBrain-screening platform to identify the long-term effects of FDA-approved drugs on the pathophysiological response to neurotoxic stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will yield a platform for in vitro experimentation that can inform the design of clinical trials, enabling them to be stratified based on genetic factors. This approach can also be combined with additional variables such as spaceflight to investigate multivariate interactions between genetics, drugs, and stressors. This study will build a foundation of knowledge and technology for determining an individual's probability of adverse neurological events in space and on Earth.
Task Progress:	Over the last funding period, we have made several advances that position us to extend and test the longevity and functionality of the miBrain. These advances include the optimization of feeding and extracellular matrices. We have validated that our improved culture conditions maintain vascular and neuronal networks for at least two months. Experiments to verify miBrain integrity at longer time points are underway. In parallel, we optimized live imaging parameters for the miBrain and developed a panel of fluorescent genetically encoded reporters to quantify sub-cellular dynamics, including mitochondrial and lysosomal function and cell proliferation. To optimize feeding methods for the miBrain, we compared three approaches: static cultures, miBrains grown in suspension in spinner flasks, and culturing miBrains on permeable transwell inserts that allow media to contact both surfaces of the miBrain. We evaluated each culture method's ability to maintain neuronal and vascular networks as well as permit live imaging. At 2 months, culturing miBrains in spinner flasks led to spherical miBrains that were very dense. On the spinner flask, miBrains live imaging was possible on the surface. However, it was not possible to visualize the interior of the cultures without sectioning the miBrains. Cryosectioning revealed that peripheral regions adjacent to the surface (10 - 50 μM) of miBrains grown in spinner flasks had vascular and neuronal networks. However, many miBrains cultured in the spinner flasks exhibited necrotic cores, suggesting poor perfusion of nutrients and oxygen. We next compared static miBrains cultured on glass bottom chambers to miBrains cultured on optically clear transwell membrains. Transwell miBrains exhibited significantly more vascular and neuronal networks than glass-bottom chambered miBrains. In the transwells, live-imaging quality was comparable to glass bottom chambers. Therefore, we moved forward with culturing miBrains on transwells. We also investigated the use of synthetic extracellular matrix (ECM) mimetics. To do this, we developed a novel, photo-cross-linked, polyethylene glycol (PEG)-based hydrogel functionalized with arginine-glycine-aspartate (RGD) peptide. We titrated the stiffness and pore size of the hydrogel to identify optimal ranges for culturing miBrains and human induced pluripotent stem cells (iPSC)-derived neuronal cells. We then compared miBrains cultured in PEG-based hydrogels to miBrains cultured in commercially available ECMs, such as Matrigel and Geltrex. At this time, both Matrigel and Geltrex outperform our PEG-based hydrogel concerning their ability to reproducibly support miBrains with robust vascular and neuronal networks. We are continuing to optimize the hydrogels, but are also moving forward with testing longevity using Geltrex, given its superior performance. In parallel with optimizing miBrain culture conditions, we have developed a panel of fluorescent reporters for live cell imaging of sub-cellular dynamics. Quantitative visualization of mitochondrial, lysosomal, and nuclear dynamics is crucial to better understand how these organelles are dysregulated during aging, in response to neurotoxic stress, and during pathogenesis. Genetically encoded fluorescent markers are a powerful tool to visualize subcellular organelles and

	compartments in real time without disrupting cellular processes. To achieve this, we constructed a toolbox of genetically encoded fluorescent markers, introduced into cell cultures by lentiviral transduction, which can be used to interrogate changes in proliferation and lysosomal and mitochondrial function in cell models of disease pathogenesis and progression. The fluorophores in this toolbox are designed to specifically localize to the desired organelle and illuminate information on localization, distribution, and number, as well as organelle turnover and intracellular oxidative stress. We have validated the functionality of the live fluorescent markers in detecting changes related to neurodegeneration. The fluorophores detect and quantify chemical and genetic mediated changes to cellular proliferation, lysosomal distribution, mitochondrial transport and turnover, and reactive oxygen species. The fluorescent toolbox was designed to be modular, such that it can be expanded to include existing or new fluorophores and targeted to any subcellular structure or cell type, greatly expanding the potential for genetically encoded fluorescent markers to probe any cell-model system or subcellular network in live cell cultures. We have demonstrated that this powerful set of live-cell fluorescent markers will provide new insight into the subcellular mechanisms that underlie neurotoxic stress responses that could be encountered in microgravity or space. This system can be applied to a diverse range of cellular models of disease to glean an understanding of the underlying mechanisms that promote neurological and cognitive degeneration. We have integrated many of the live imaging reporters into the miBrain. We are developing high-content image analysis and machine learning algorithm pipelines to quantitatively analyze images from the miBrain. Over the next year, we plan to generate panels of multiplexed live imaging assays to systematically interrogate cellular function and degeneration in the miBrain. We are also continuously testing the longevity of the miBrain and iteratively improving the culture conditions to increase the robustness and quality of the miBrain system. Ultimately, our efforts will deliver an in vitro human brain tissue that is a reliable proxy of the human brain’s response to neurotoxic stress.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Other Journals or Periodicals	Stanton AE, Bubnys A, Agbas E, James B, Park DS, Jiang A, Pinals RL, Truong N, Loon A, Staab C, Liu L, Cerit O, Wen H-L, Kellis M, Blanchard JW, Langer RS, Tsai L-H. "Engineered 3D immuno-glial-neurovascular human brain model." bioRxiv preprint server. Posted August 17, 2023. https://doi.org/10.1101/2023.08.15.553453 , Aug-2023